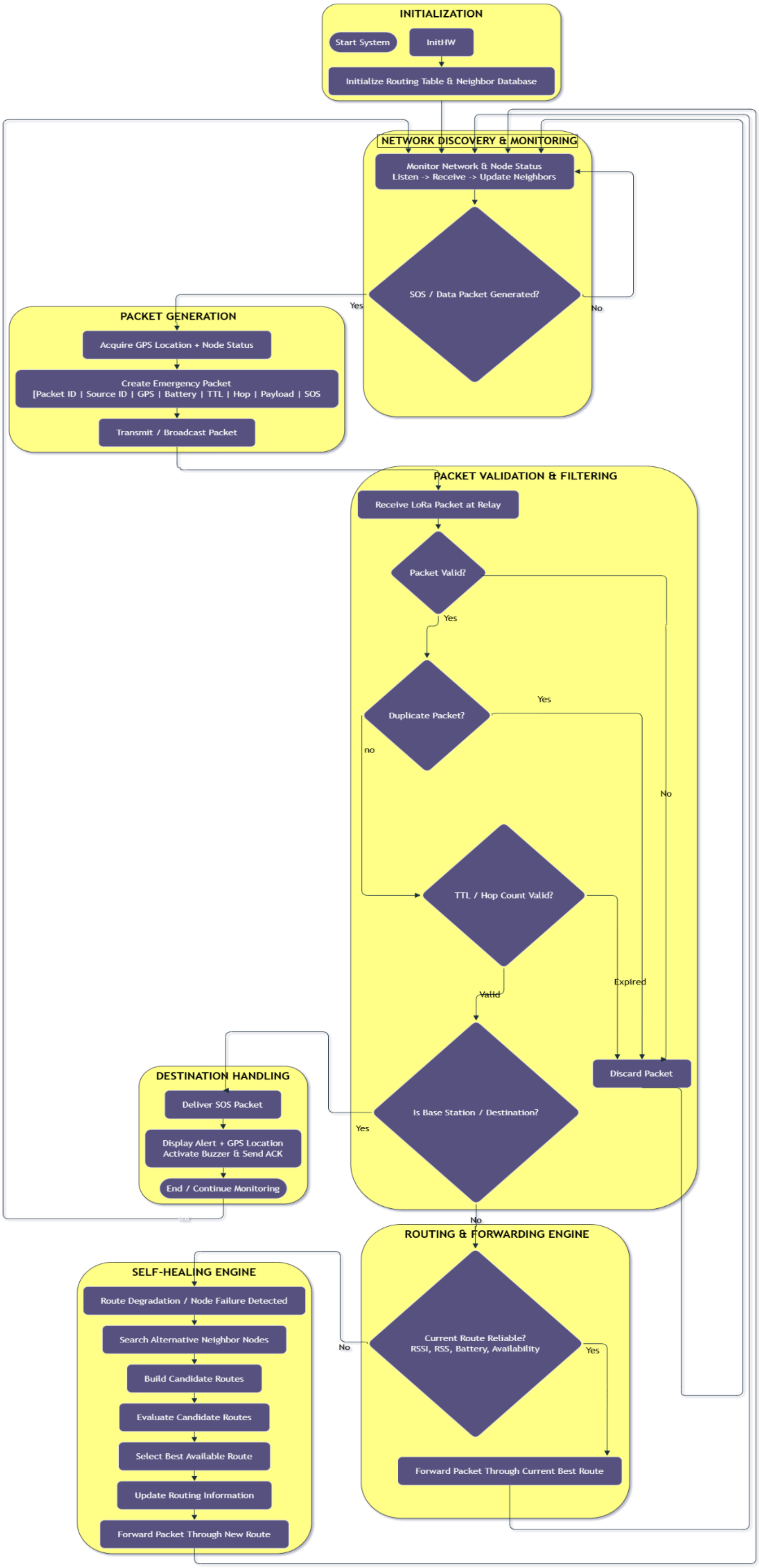
FLOW CHART:


INITIALIZATION
Start System
InitHW
Initialize Routing Table & Neighbor Database
NETWORK DISCOVERY & MONITORING
Monitor Network & Node Status
Listen -> Receive -> Update Neighbors
SOS / Data Packet Generated?
Yes
No
PACKET GENERATION
Acquire GPS Location + Node Status
Create Emergency Packet
[Packet ID | Source ID | GPS | Battery | TTL | Hop | Payload | SOS
Transmit / Broadcast Packet
PACKET VALIDATION & FILTERING
Receive LoRa Packet at Relay
Packet Valid?
Yes
No
Duplicate Packet?
Yes
no
TTL / Hop Count Valid?
Valid
Expired
Discard Packet
Is Base Station / Destination?
Yes
No
DESTINATION HANDLING
Deliver SOS Packet
Display Alert + GPS Location
Activate Buzzer & Send ACK
End / Continue Monitoring
ROUTING & FORWARDING ENGINE
Current Route Reliable?
RSSI, RSS, Battery, Availability
Yes
No
Forward Packet Through Current Best Route
SELF-HEALING ENGINE
Route Degradation / Node Failure Detected
Search Alternative Neighbor Nodes
Build Candidate Routes
Evaluate Candidate Routes
Select Best Available Route
Update Routing Information
Forward Packet Through New Route

ALGORITHM:

1. **START**
2. Initialize MCU, LoRa, GPS, SOS input, display, and network tables.
3. Check **LoRa initialization**.
   - If **NO**, retry initialization.
   - If **YES**, continue.
4. Continuously monitor nodes, links, and SOS input.
5. If **SOS = YES**, acquire GPS data.
6. Form the packet with **Source ID, Packet ID, GPS, Battery, SOS, TTL, Hop Count, and Destination**.
7. Discover and evaluate available neighbouring nodes.
8. Select a suitable **next hop**.
   - If no next hop is available, wait/retry neighbour discovery.
9. Check **TTL**.
   - If expired, **drop packet**.
   - Otherwise, decrement TTL and increment Hop Count.
10. Transmit the packet to the selected next hop.
11. At the receiving node, validate the packet.
    If invalid, **drop packet**.
12. Check for duplicate using **Source ID + Packet ID**.
    If duplicate, **drop packet**.
13. Check whether the **destination is reached**.
    - ✓ If **YES**, deliver packet to the Base Station.
    - ✓ If **NO**, continue forwarding.
14. Check node/link status.
    If the current next hop has failed, mark it **unavailable**.
15. Discover an alternative neighbour and **recalculate the next hop**.
16. Forward the packet through the new route.
17. At the Base Station, display emergency information and activate the alert.
18. Return to **continuous monitoring**.
19. **STOP / Repeat continuously**.